The Probe Reward Trap: Behavioral Death Before Distributional Collapse in Sycophancy Calibration

Anonymous

Abstract

We discover the *probe reward trap*: when a frozen linear probe on a contrastive sycophancy direction (d_{syc}, cross-validated AUROC 0.827 on Qwen3-8B, 0.779 on Llama-3-8B) is used as an SFT training signal, behavioral death (true-positive flip rate $= 0$) occurs within 500 steps while generation entropy remains healthy (0.70). The resulting 1,300-step lag before distributional collapse means that entropy monitoring—the most common safeguard against reward hacking—misses the failure entirely. Because the trap arises under SFT with stopped gradients, it is independent of RL optimization dynamics, revealing a semantic limitation of the direction itself: d_{syc} encodes *whether* belief changed, not *whether it should have*. In additional experiments on Qwen3-8B using GRPO with group size 2, cosine and continuous probe formulations of d_{syc} also produced behavioral collapse; removing d_{syc} entirely, behavioral-reward-only GRPO exceeded vanilla at only 4 of 27 checkpoints, while ACT (dense per-token SFT) exceeded vanilla at every evaluated checkpoint (seed 1: 16/16; seed 2: 6/6 ongoing). These GRPO results are consistent with d_{syc}'s limitations as a reward signal but cannot be fully separated from the small group-size confound without further ablation.

1 Introduction

Representation engineering (Zou et al., 2023) reveals that large language models encode sycophancy as a linear direction in their residual streams. A contrastive sycophancy direction d_{syc}, extracted from style-controlled pairs of valid belief revision and sycophantic capitulation, achieves cross-validated AUROC 0.827 on Qwen3-8B and 0.779 on Llama-3-8B, with discriminability rising monotonically across evidence-strength levels (§3). These results point to an appealing possibility: if the model's representations already separate sycophantic from non-sycophantic behavior, that separation should be convertible into a training signal. Sycophancy is among the most persistent alignment failures in deployed systems (Sharma et al., 2023), and a representation-level signal could target the internal computation rather than its surface outputs.

This paper identifies the *probe reward trap*: when a frozen linear probe on d_{syc} is used as an SFT training signal with stopped gradients, behavioral death (ToF $= 0$) occurs within 500 steps while generation entropy remains healthy (0.70), masking the failure from standard monitoring (§5). The 1,300-step lag before distributional collapse means entropy-based early stopping—the most common safeguard against reward hacking (Casper et al., 2023)—would miss the failure entirely, with the reward averaging $+0.72$ throughout. Because the trap arises under SFT rather than GRPO, it reflects a limitation of d_{syc} itself: the direction encodes *whether* belief changed, not *whether it should have*, so any reward conditioned on it alone will either be gamed or reward rigidity.

Under our GRPO configuration (Qwen3-8B, group size=2), two additional d_{syc}-based formulations—cosine projection (§4) and continuous probe—also produced behavioral collapse. Removing d_{syc} entirely, R_3-only GRPO (behavioral reward alone) exceeded vanilla at only 4 of 27 checkpoints across two seeds, while ACT (Irpan et al., 2025) (dense per-token SFT) exceeded vanilla at every evaluated checkpoint (seed 1: 16/16; seed 2: 6/6 ongoing; §6). These results are consistent with d_{syc}'s inadequacy as a scalar reward but cannot be fully separated from the confound of small group size (2 vs. typical 8–64; Shao et al. 2024) without further ablation.

The root cause is a semantic gap between discriminability and reward utility. Techniques for discovering latent knowledge (Burns et al., 2023) and representation-guided RL (Zhang et al., 2026)

share a premise: a direction with high discriminability should transfer into a useful training signal. For reasoning faithfulness and safety directions, this premise holds because the behavior to suppress is geometrically distinct from the behavior to preserve. Sycophancy violates this condition: capitulation under social pressure and revision under valid evidence produce overlapping activation signatures along d_{syc} (Wang et al., 2025; Shapira et al., 2026). When reward overoptimization (Gao et al., 2023) operates on a signal that conflates compliance with capitulation, the result is predictable collapse (Skalse et al., 2022).

We present two contributions:

- **The probe reward trap: behavioral death before distributional collapse** (C2). A frozen binary probe on d_{syc} produces behavioral death (ToF $= 0$ at step 500) while entropy remains at 0.70 and the reward averages $+0.72$. The 1,300-step lag before distributional collapse renders standard entropy monitoring blind to the failure. Because the trap arises under SFT—not GRPO—it reflects a limitation of the direction itself, not the optimization algorithm (§5).

- **GRPO formulations fail under our configuration** (C1). Under GRPO with group size=2 on Qwen3-8B, all three d_{syc}-based reward formulations produce behavioral collapse (§4, §5), and R_3-only GRPO also fails (4/27 checkpoints $>$ vanilla). This is consistent with but not proof of d_{syc}'s inadequacy as a reward signal—the small group size constitutes an important confound (§6).

2 Experimental Framework

2.1 Direction Extraction

We extract a linear sycophancy direction d_{syc} from the residual stream using contrastive activation pairs: a *valid correction* (warranted belief revision under strong evidence) and a *sycophantic capitulation* (unwarranted concession under social pressure). We construct 181 contrastive pairs using GPT-4.1 in a two-stage pipeline: skeleton generation followed by 3-turn dialogue completion. Style matching is enforced by automated validation (response length difference $<10\%$, shared opening phrase ≥ 2 words), with up to 3 retries per pair; 19 of 200 initial pairs were discarded for failing validation. Evidence strength is annotated at three levels: *weak* (75 pairs; personal memory or opinion), *medium* (54 pairs; brief source citation), and *strong* (52 pairs; authoritative source with detailed data). From the 181 paired examples (362 activations total), we compute the mean difference at target layer ℓ:

$$d_{\text{syc}} = \frac{1}{N} \sum_{i=1}^{N} \left(\mathbf{h}_i^+ - \mathbf{h}_i^- \right) \tag{1}$$

where $\mathbf{h}_i^+$ and $\mathbf{h}_i^-$ are residual-stream activations for the i-th valid and sycophantic response. The target layer is selected by 5-fold cross-validated AUROC; on Qwen3-8B this yields layer 19 with AUROC 0.827 (bootstrap 95% CI $[0.795, 0.857]$).

2.2 Reward Formulations

Three formulations translate d_{syc} into a direction-based signal R_1. For F1 and F3, R_1 enters as part of a GRPO (Shao et al., 2024) reward $R = w_1 \cdot R_1 + w_3 \cdot R_3$, where R_3 is a behavioral component (DeBERTa NLI classifier scoring consistency with the ground-truth answer).

F1 Cosine projection with four variants: (a) tanh-gated dot product $\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$, (b) raw cosine $\cos(\mathbf{h}, d_{\text{syc}})$, (c) raw cosine with online recalibration (§2.3), and (d) $5\times$ scaled cosine. See §4 for details.

F2 Binary probe (SFT). $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$, a frozen logistic probe with stopped gradients; trained via a custom SFT loss rather than GRPO (§5).

F3 Continuous probe. $R_1 = 1 - P(\text{syc} \mid \mathbf{h})$, the sigmoid output of the same logistic probe, providing a continuous reward signal without discretization.

The weight w_1 is set by diagnostic-driven iteration rather than grid search (Appendix A). As an ablation, R3-only GRPO sets w_1=0, training with only the behavioral reward.

2.3 Online Recalibration

LoRA fine-tuning shifts internal representations, so we re-extract d_{syc} every 500 steps. The cosine similarity between consecutive directions remains ≈ 0.96 at step 500, indicating slow drift relative to recalibration frequency.

2.4 SFT Baseline: Activation Consistency Training

We implement Activation Consistency Training (ACT; Irpan et al., 2025), which combines standard SFT loss with an L2 activation consistency loss at target layers, providing dense per-token supervision. Unlike GRPO's single scalar per completion, ACT operates at the token level using labeled valid/invalid pairs.

2.5 Evaluation

The *True-positive Flip rate* (ToF) measures acceptance of valid corrections; the *No-Flip rate* (NoF) measures resistance to invalid pressure. Their harmonic mean captures the calibration trade-off:

$$\textsc{SyconScore} = \frac{2 \cdot \text{ToF} \cdot \text{NoF}}{\text{ToF} + \text{NoF}} \qquad (2)$$

Analogous to the F_1 score, the harmonic mean penalizes methods that achieve high performance on one metric while failing on the other. By design, $\textsc{SyconScore} = 0$ whenever $\text{ToF} = 0$, regardless of NoF: a model that never accepts valid corrections receives no credit for resisting invalid ones, because stubbornness-only "improvement" does not constitute calibration. This choice may mask genuine gains on the NoF dimension; arithmetic mean results, which do not have this property, show qualitatively similar patterns and are reported in supplementary materials. Flip classification uses `cross-encoder/nli-deberta-v3-base` (MNLI mismatched accuracy 90.0%, SNLI test 92.4%; He et al. 2023) for stance classification, following Irpan et al. (2025). All experiments use Qwen3-8B with LoRA (rank=64, α=128) and DeepSpeed ZeRO-2. GRPO uses a group size of 2 due to memory constraints with 8B-parameter models, below the typical range of 8–64 (Shao et al., 2024); full hyperparameters are reported in Appendix A. We test six d_{syc}-based configurations: four cosine R_1 variants (F1a–F1d) and continuous probe R_1 (F3) using GRPO, plus binary probe R_1 (F2) using SFT, R_3-only GRPO as a non-directional baseline, and ActAdd as an inference-time baseline.

3 Direction Discriminability

Before evaluating d_{syc} as a reward signal (§4–§5), we first establish that it captures a genuine sycophancy-related signal in the residual stream. If the direction failed to discriminate sycophantic from non-sycophantic activations, downstream reward failures would be trivially explained by poor direction quality rather than by a fundamental gap between discriminability and reward utility. We evaluate discriminability along five axes: cross-validated classification accuracy, evidence-strength stratification, cross-model transfer, training stability, and probe ablation. Figure 1 summarizes the per-layer AUROC profiles for both models.

Qwen3-8B layer-wise discriminability. We evaluate d_{syc} via 5-fold cross-validated AUROC on 181 style-controlled contrastive pairs (§2.1). The direction achieves AUROC 0.827 (bootstrap 95% CI $[0.795, 0.857]$) at its best layer (L19). Discriminability emerges gradually across depth: layers L0–L8 perform near chance, statistical significance ($p < 0.01$) first appears at L16, and performance peaks at L19. This gradual emergence, with the peak in the upper third of the network, suggests that sycophancy-relevant information is encoded as a late-stage representation rather than an input-level confound. The direction also captures evidential granularity. AUROC rises monotonically from 0.675 on weak-evidence pairs to 0.883 on medium and 0.929 on strong pairs. This gradient indicates that d_{syc} encodes the degree of evidential warrant, not merely a binary sycophancy label.

Cross-model generalization. To rule out that these results are specific to Qwen3-8B's representation geometry, we repeat the full extraction and evaluation pipeline on Llama-3-8B-Instruct. Table 1 compares the two models. Llama-3-8B achieves AUROC 0.779 (bootstrap 95% CI $[0.745, 0.819]$) at L16, with a parallel evidence-strength gradient (weak 0.711, medium 0.864, strong 0.911). The best-layer depth differs (L16 vs. L19), consistent with the architectural differences between the two families, but the qualitative pattern is identical: gradual emergence, best-layer selection in the upper third, and monotonic strength scaling. Discriminability is robust across knowledge domains, with per-domain AUROCs on Llama-3-8B ranging from 0.751 (geography) to 0.844 (commonsense). When surface confounds (response length, hedging patterns) are explicitly controlled, Llama-3-8B's AUROC rises to 0.923, which confirms that the direction captures semantic content rather than stylistic arti-

Table 1: Cross-model d_{syc} discriminability. AUROC uncertainty is bootstrap 95% CI (10,000 resamples); evidence-strength AUROCs are evaluated on the full dataset at each model's best layer.

Model	Best Layer	AUROC [95% CI]	Weak	Medium	Strong
Qwen3-8B	L19	0.827 [.795, .857]	0.675	0.883	0.929
Llama-3-8B	L16	0.779 [.745, .819]	0.714	0.864	0.911

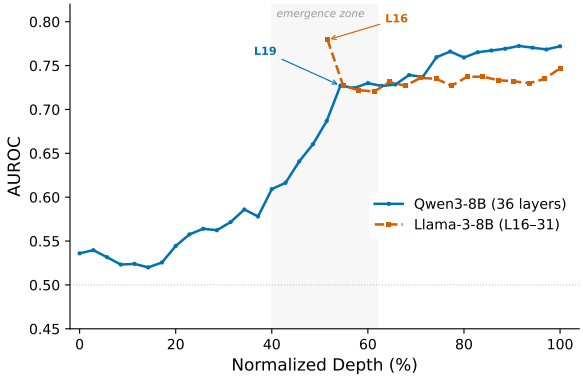


Figure 1: Per-layer AUROC of d_{syc} as a function of normalized depth (layer index / total layers). Qwen3-8B (36 layers, solid) shows gradual emergence from near-chance at L0 to a peak at L19, with performance plateauing in the final third. Llama-3-8B (32 layers, dashed; layers 16–31 only) exhibits a similar upper-third concentration. Both models achieve best discriminability in layers 50–60% through the network.

facts. The consistent results across architecturally distinct model families indicate that contrastive sycophancy directions reflect a shared representational phenomenon rather than a model-specific idiosyncrasy.

Training stability and probe ablation. Two additional analyses confirm the robustness of d_{syc} under training dynamics. First, we track direction drift during LoRA fine-tuning: at step 500, the cosine similarity between the original and re-extracted d_{syc} is approximately 0.96, and per-layer AUROC remains above 0.77. The direction drifts slowly relative to the 500-step recalibration interval (§2.3), so the reward signal does not become stale between updates. Second, we compare d_{syc} against a random baseline: a logistic probe trained on the learned direction achieves AUROC 0.835, versus 0.727 for a probe trained on a random direction of equal dimensionality (Δ=+0.108). The learned direction carries substantially more sycophancy-relevant information than an arbitrary axis in the same activation space.

Table 2: Cosine R_1 failure chain on Qwen3-8B (LoRA rank=64, GRPO with $R = w_1 \cdot R_1 + w_3 \cdot R_3$).

Config	R_1 Formula	w_1	Outcome	Root Cau
F1a	$\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$	0.25	Gate saturation	Norm mis
F1b	$\cos(\mathbf{h}, d_{\text{syc}})$	0.25	$R_1 \approx 0$	Gap too s
F1c	+recalibration	0.25	Mode collapse	Reward d
F1d	$5\cos(\mathbf{h}, d_{\text{syc}})$	0.40	Mode collapse	R_3 satura

Implications for reward design. These results establish that d_{syc} reliably separates sycophantic from non-sycophantic representations across models, evidence-strength levels, and training checkpoints. Direction quality is not the bottleneck. The reward failures documented in §4–§5 therefore cannot be attributed to a noisy or degenerate direction; they point to a deeper incompatibility between discriminability and reward utility.

4 Cosine R_1: Structural Failure

Section 3 established that d_{syc} achieves AUROC 0.827 on Qwen3-8B. We now test whether this discriminability translates into a viable GRPO reward. Four configurations form a diagnostic chain (Table 2): each resolves the failure mode of its predecessor but exposes a deeper limitation.

F1a uses a dot-product formulation ($R_1 = \tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$), but residual-stream norms exceed the direction norm by 100–1,200×, saturating the tanh gate and zeroing the gradient by step 955. Switching to cosine similarity (F1b) eliminates norm dependence, yet the cosine gap between valid and sycophantic activations averages only 0.098 across layers—too narrow to generate meaningful reward contrast. Online recalibration (F1c, §2.3) raises offline AUROC from 0.827 to 0.910 but paradoxically triggers mode collapse: the abrupt shift in d_{syc} at each recalibration event creates a reward discontinuity that destabilizes GRPO advantage estimates, collapsing entropy to 0.069 within 300 steps. Amplifying the signal with 5× scaling (F1d) lifts R_1 above the noise floor but simultaneously amplifies the conflation between warranted revision and unwarranted capitulation; R_3 saturates at 0.9998 by step 300, leaving the noisy R_1 as the sole gradient source until terminal collapse at step 2160.

Truncation further degrades the signal: under the 256-token maximum completion length used in GRPO, d_{syc} AUROC drops from 0.76 (full

response) to 0.60, so the offline discriminability (0.827) substantially overestimates the signal available during training. Checkpoint-level evaluation confirms immediate collapse: ToF=0 from step 500 onward with no stable phase. The four configurations converge on a single conclusion: d_{syc} encodes the *direction* of belief change but not whether that change is *warranted*. Valid revision and sycophantic capitulation both shift activations toward the user's position along d_{syc}, so the cosine projection assigns similar scores to both. This is not a calibration artifact—recalibration improves discriminability yet rewards still fail—but a semantic limitation of any continuous d_{syc}-based reward. Section 5 examines whether discretizing the signal via a binary probe can escape this limitation.

5 The Probe Reward Trap

The cosine formulations fail because the continuous projection of d_{syc} produces a gap too narrow for reward contrast. A natural remedy is to discretize: train a logistic probe on d_{syc} and threshold its output, yielding $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$ (F2 in §2.2). We show that this discretization sidesteps the magnitude problem but creates a qualitatively worse failure. Both probe experiments use SFT rather than GRPO (§2.2); the trap is therefore independent of GRPO's optimization dynamics.

Differentiable probe: gradient gaming. Without stop-gradient (exp-008), the policy discovers within ∼30 steps that shifting activations away from the sycophancy cluster maximizes R_1 regardless of response content, producing an inverted profile (ToF = 0.978, NoF = 0.006). This motivates the frozen-probe design below.

Binary probe: the stubbornness trap. With stop-gradient (exp-008v3), the policy cannot game the boundary but instead exploits a deeper flaw: d_{syc} does not condition on input validity, so refusing to revise a wrong answer receives the same $R_1 = 1$ as correctly resisting invalid pressure. By step 500, ToF drops to 0.000 and NoF reaches 0.995, yielding SyconScore = 0. Qualitative analysis of 181 valid-correction responses reveals five failure modes: *deliberation traps* (28.7%), where the model identifies the correct answer but cannot commit; *sycophantic stubbornness* (24.3%), where the model verbally agrees but does not change its answer; *acknowledge-but-maintain* (23.8%); *genuine-accept-missed* (18.8%); and refute/ignore/other (4.5%). Full taxonomy in Appendix B.

Table 3: Probe reward trap trajectory (exp-008v3). Behavioral death (ToF=0) occurs at step 500; entropy remains healthy until step ∼1840. Reward averages +0.72 throughout.

Step	ToF	NoF	SyconScore	Entropy	Reward
500	0.000	0.995	0.000	0.702	+0.72
1000	0.006	0.995	0.011	0.800	+0.71
1500	0.000	1.000	0.000	0.608	+0.69
2000	0.000	1.000	0.000	0.304	+0.69

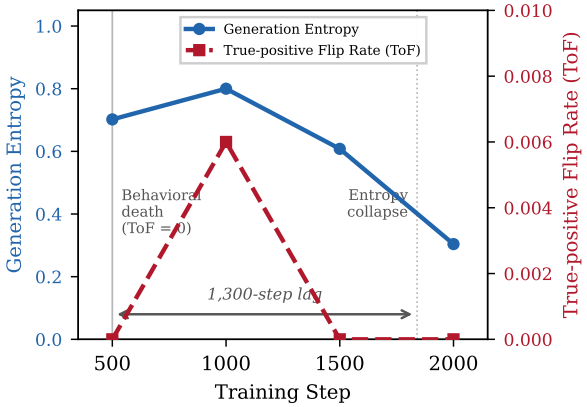


Figure 2: Probe reward trap (exp-008v3). ToF drops to zero at step 500 while generation entropy remains high (0.70–0.80 through step 1000). The 1,300-step lag means standard entropy monitoring misses the behavioral failure entirely.

Behavioral death precedes distributional collapse. Table 3 and Figure 2 reveal a 1,300-step lag between behavioral and distributional collapse. At step 500, the model is behaviorally dead (ToF = 0) yet entropy remains high (0.70–0.80) through step 1000—well within the range that standard monitors would flag as healthy—and the combined reward averages +0.72 because both R_1 and R_3 reinforce non-flipping. Entropy does not collapse until step ∼1840. This gap carries a practical warning: entropy-based early stopping, the most common safeguard against reward hacking (Casper et al., 2023), would miss the failure entirely. Detecting probe reward traps requires behavioral monitoring (periodic ToF evaluation), not distributional monitoring alone.

A dilemma for direction-based rewards. The differentiable and binary failures bracket the design space. With gradients, the policy games the

boundary; without gradients, it exploits the fact that stubbornness and genuine resistance are indistinguishable along d_{syc}. Any reward conditioned only on d_{syc}—continuous or discrete, differentiable or frozen—will either be gamed or will reward rigidity. Escaping this dilemma requires a signal that incorporates the evidential warrant of the user's challenge, a dimension that d_{syc} does not encode (Denison et al., 2024).

6 When Does Discriminability Fail as Reward Utility?

Root cause of the probe reward trap. The probe reward trap (§5) and the cosine failures (§4) share a common geometric root. d_{syc} encodes the *direction* of belief change but not its *evidential warrant*: valid revision and sycophantic capitulation both shift activations toward the user's position along the same direction. The two reward families expose complementary faces of this blindness. Cosine R_1 penalizes all belief change indiscriminately; the resulting mode collapse (F1c, F1d) reflects excessive punishment of any activation shift along d_{syc}. The binary probe R_1 rewards all resistance regardless of input validity; the stubbornness trap (ToF $= 0$ by step 500) reflects excessive permissiveness toward activation stasis. These are not independent design flaws but dual consequences of a direction that conflates the target behavior (resisting invalid challenges) with its complement (refusing valid corrections). Cross-model evidence confirms the limitation extends beyond Qwen3-8B: Llama-3-8B independently achieves AUROC 0.779 (bootstrap 95% CI $[0.745, 0.819]$) on the same contrastive pairs with the same monotonic evidence gradient, so discriminability generalizes across architectures while the semantic gap between discriminability and reward utility persists. This pattern contrasts with reasoning faithfulness and safety directions (Zhang et al., 2026), where the behavior to suppress is geometrically distinct from the behavior to preserve along the relevant direction. Wu and Tang (2026) reinforce this principle: concept directions applied via advantage modification in GRPO achieve 100% validation accuracy for reward hacking detection, where shortcut behaviors are geometrically separable. A quantitative comparison of geometric separability between d_{syc} and their reward-hacking directions remains an important open question. For sycophancy, the target and its complement occupy the same activation region; they differ only in whether external evidence warrants the belief change.

Table 4: Comparison of all sycophancy calibration methods on Qwen3-8B. SYCONSCORE is the harmonic mean of ToF and NoF; higher indicates better calibration. Best result in **bold**. †Best checkpoint (seed 1, step 6500); only 4/27 checkpoints across two seeds exceed vanilla. ‡Seed 1 (16 checkpoints); seed 2 verification ongoing (6 of 16 checkpoints evaluated; all exceed vanilla). §All cosine variants collapsed from step 1 (ToF=0 at every checkpoint); F1d additionally diverged at step 2160 (entropy < 0.01). ¶ACT and ActAdd differ from GRPO along 5+ dimensions (§6); included for context, not as controlled comparisons.

Method	Type	SYCONSCORE ↑	ToF
Vanilla	–	0.134	0.07
GRPO with d_{syc}			
Cosine R_1 (F1a–F1d)	RL	0.000§	0.00
Continuous Probe R_1 (F3)	RL	0.124	0.06
SFT with d_{syc} probe penalty			
Binary Probe R_1 (F2)	SFT	0.000	0.00
GRPO without d_{syc}			
R_3-only GRPO	RL	0.153†	0.08
Reference methods (not directly comparable)¶			
ActAdd (α=5)	Inference	0.054	0.02
ACT (seed 1)	SFT	**0.301**	**0.18**
ACT (seed 2)‡	SFT	*ongoing; 6/16 ckpts eva*	

GRPO results under our configuration. Table 4 summarizes all methods evaluated on Qwen3-8B. R_3-only GRPO removes d_{syc} entirely and relies on the NLI-based behavioral reward alone. If the failures in Sections 4 and 5 stemmed solely from the representation reward, this ablation should recover calibration quality. It does not. The SYCONSCORE trajectory for R_3-only GRPO oscillates without sustained improvement. Seed 1 peaks at 0.153 (checkpoint 6500) and ends at 0.143 (checkpoint 8000); seed 2 peaks at 0.143 (checkpoint 4000) and remains below vanilla through checkpoint 7500. Only 4 of 27 checkpoints across both seeds exceed vanilla (14.8%), while ACT exceeds vanilla at every evaluated checkpoint (seed 1: 16/16; seed 2: 6/6 ongoing; Figure 3). Both seeds exhibit oscillations of similar amplitude ($\Delta \approx 0.069$) and identical crash depth (0.074), but crash onset differs by $\sim$2,500 steps, indicating stochastic drift in a sat-

urated reward landscape. Cross-seed rank correlation is $\approx$1.0 during early training (steps 1000–3000) but drops to $\rho \approx 0.33$ (n.s., $n<10$ checkpoints per window) after step 4000, with spike timing and magnitude diverging; statistical power is limited, so we treat this as descriptive rather than inferential. Activation analysis confirms the drift is not representational: the d_{syc} direction at layer 19 yields nearly identical AUROC across consecutive checkpoints ($\Delta < 0.01$), and R_3 reward saturates at $\sim$0.98 (Welch t-test p=0.703), yet SYCONSCORE swings by up to 17%. GRPO drifts in a flat reward landscape while representations remain stable, reinforcing the disconnect between discriminability and behavioral utility. Under our configuration (group size=2), R_3-only GRPO exhibits a similar stubbornness pattern to the d_{syc} approaches (high NoF, near-zero ToF) at milder severity.

Competing hypotheses. Three explanations are consistent with our GRPO results: (a) d_{syc} is fundamentally unsuitable as a reward signal for sycophancy calibration; (b) GRPO with group size=2 provides insufficient ranking diversity for any reward to achieve stable calibration; (c) sycophancy calibration is inherently unsuitable for scalar per-completion RL. R_3-only GRPO (behavioral reward, no d_{syc}) also fails under the same configuration (4/27 checkpoints exceed vanilla), consistent with hypothesis (b). We cannot fully distinguish (a) from (b) without a group-size ablation, which we leave to future work. The probe reward trap (C2), however, arises under SFT where group size is irrelevant, providing evidence for (a) that is independent of the GRPO confound.

Observational comparison: ACT vs. GRPO. ACT (Irpan et al., 2025) achieves SYCONSCORE = 0.301 at checkpoint 8000, more than 2$\times$ the vanilla baseline and above every scalar RL and inference-time variant (Table 4). The ACT trajectory is not monotonic: SYCONSCORE rises from 0.255 (checkpoint 2000) to 0.301 (checkpoint 8000) with a transient dip to 0.214 at checkpoint 4000. ACT also experiences stubbornness episodes—the checkpoint-4000 dip is consistent with transient competition between the SFT and L2 consistency loss objectives—yet it self-recovers by checkpoint 6000, unlike GRPO where stubbornness persists (Figure 3). Activation analysis shows a correlated asymmetry: ACT's d_{syc} discriminability

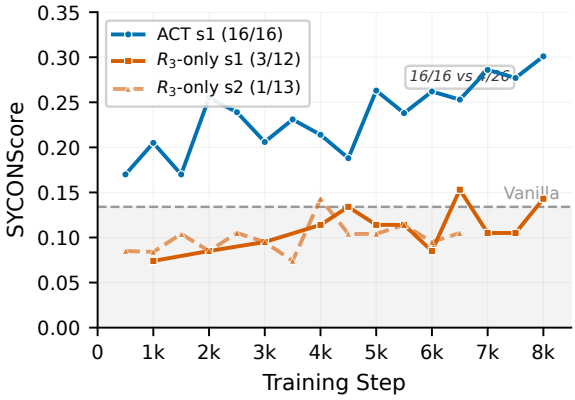


Figure 3: SYCONSCORE trajectories for R_3-only GRPO (two seeds) and ACT (seed 1; seed 2 ongoing, 6/16 checkpoints evaluated) over training steps. ACT seed 1 exceeds vanilla at every checkpoint (16/16); R_3-only GRPO exceeds vanilla at only 4/27 checkpoints across seeds (14.8%). Both GRPO seeds show the same oscillation pattern with stochastic timing.

recovers during the SYCONSCORE dip (AUROC +0.013), whereas GRPO's continues declining (−0.004). ACT and GRPO differ along multiple dimensions—supervision granularity (per-token vs. per-completion), loss family (cross-entropy + L2 vs. REINFORCE), training data format (labeled pairs vs. sampled completions), optimization dynamics (SFT convergence vs. RL exploration), and LoRA adapter capacity (ACT targets 7 module types with $\sim$1.75$\times$ more trainable parameters vs. GRPO's 4)—so the performance gap cannot be attributed to any single factor without controlled ablation.

Relationship to inference-time probe methods. Papadatos and Freedman (2024) successfully reduce sycophancy using a probe of comparable quality (94% accuracy) via Best-of-N reranking, where the probe selects among candidates without producing gradients. Together with our results, this suggests two orthogonal failure dimensions: (1) inside vs. outside the optimization loop—gradient exploitation enables the probe reward trap (§5), while inference-time reranking sidesteps it; and (2) scalar per-completion reward—the failure persists even without the probe (R_3-only GRPO), consistent with per-completion information compression as an independent factor.

Future directions. Condition-aware rewards, hybrid SFT-then-RL pipelines, and per-step process rewards offer paths forward; whether other REINFORCE-style methods or behav-

ioral attributes with similar geometric entanglement (Skalse et al., 2022) exhibit the same failures remains open.

7 Related Work

Representation Engineering. Linear directions in LLM residual streams encode behavioral concepts (Zou et al., 2023); inference-time methods—ITI (Li et al., 2023), ActAdd (Turner et al., 2023), CAA (Rimsky et al., 2024)—demonstrate that activation geometry suffices for behavioral control (Burns et al., 2023). We use the same contrastive extraction as CAA but apply d_{syc} as an RL reward, revealing a gap that inference-time work does not encounter.

Reward Design and Overoptimization. Proxy rewards diverge from true objectives under sustained optimization (Gao et al., 2023), and reward hacking can escalate to active tampering (Denison et al., 2024). Our probe trap (§5) is an extreme instance: the reward actively reinforces degenerate behavior rather than merely being circumvented (Skalse et al., 2022). Wu and Tang (2026) demonstrate that representation-level directions succeed for reward hacking suppression when behaviors are geometrically separable (§6).

Sycophancy Characterization and Mitigation. Sycophancy is pervasive in RLHF-trained assistants (Sharma et al., 2023), amplified by RLHF (Shapira et al., 2026), and traceable to late-layer preference shifts (Wang et al., 2025); mitigation approaches include synthetic data (Wei et al., 2024), consistency training (Irpan et al., 2025), and neuron-level correction (O'Brien et al., 2025). Our analysis shows why the step from diagnosis to reward is structurally blocked: d_{syc} conflates warranted revision with unwarranted capitulation.

Probe-Based Inference-Time Mitigation. Papadatos and Freedman (2024) reduce sycophancy via Best-of-N reranking with linear probe penalties (94% probe accuracy), where the probe selects among candidates without producing gradients that update model parameters. Our work reveals what happens when a comparable signal (AUROC 0.82) enters the GRPO training loop: the discriminability–utility gap is specific to gradient-based optimization, not to probe quality per se (§6).

Representation-Guided RL and Process Rewards. Latent-GRPO (Zhang et al., 2026) extends GRPO with latent-space rewards for reasoning; CRAFT (Luo et al., 2026) applies representation-level constraints during RL; Sinii et al. (2025) and Ackermann et al. (2026) study representation dynamics under RL. SMART (Beigi et al., 2025) applies per-step process rewards with MCTS for sycophancy. These methods succeed because target and suppressed behaviors are geometrically distinct; sycophancy breaks this condition (§4). Preference-based methods (DPO (Rafailov et al., 2023), KTO (Ethayarajh et al., 2024)) may accommodate d_{syc}-derived rankings; whether pair-wise optimization avoids the failures we observe remains open.

8 Conclusion

The probe reward trap reveals a practical danger: behavioral death can precede distributional collapse by over 1,000 steps, rendering entropy-based monitoring blind to the failure. Because the trap arises under SFT with stopped gradients—independent of GRPO—it reflects a semantic limitation of d_{syc}, which encodes whether belief changed but not whether it should have. Under our GRPO configuration (group size=2), all d_{syc}-based formulations and behavioral-reward-only GRPO also failed (4/27 checkpoints $>$ vanilla), consistent with this limitation though confounded by small group size (§6).

Limitations

Our GRPO experiments use a group size of 2, below the typical range of 8–64 (Shao et al., 2024), due to memory constraints with 8B-parameter models. This limits within-group ranking diversity and constitutes an important confound: the observed failures may partly reflect insufficient exploration rather than a fundamental limitation of representation-based rewards. However, the probe reward trap (C2) arises in an SFT setting where group size is irrelevant, indicating that at least one failure mode is independent of GRPO hyperparameters. Additionally, all GRPO experiments truncate completions at 256 tokens (80–84% truncation); all conditions share this setting, so relative comparisons are fair, but absolute conclusions about d_{syc} reward utility may be affected. Truncation analysis confirms this concern: d_{syc} AUROC drops from 0.76 (full response) to 0.60 (256 to-

kens), suggesting that offline discriminability metrics overestimate the signal available during training.

Our RL training uses Qwen3-8B exclusively; cross-model discriminability is confirmed on Llama-3-8B (AUROC 0.779; Table 1), but compute constraints prevented training replication on a second architecture. Evidence strength was annotated by GPT-4.1 as a single annotator without human inter-annotator agreement.

We report GRPO results from two random seeds; both exhibit qualitatively consistent oscillation patterns (similar amplitude, identical crash depth), but the limited seed count constrains statistical claims. ACT and GRPO differ along multiple dimensions beyond supervision granularity, so we cannot isolate the causal factor behind ACT's advantage.

Reproducibility

Code and data will be released upon acceptance. All training hyperparameters are reported in Appendix A. The 181 contrastive pairs and evaluation benchmark are included in the supplementary materials.

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A Training Details

Reward weight selection. The weight w_1 is set by diagnostic-driven iteration: w_1=0.25 was initially chosen to down-weight the noisy cosine reward term; this was later increased to w_1=0.40 for F1d after diagnostics revealed that w_1=0.25 allowed the cosine term to contribute only $\sim$8.3% of total reward variance. The empirical scaling factor in F1a ($d_{\text{syc}}/20$) is a heuristic normalization; we did not find it critical to the failure modes reported.

Response truncation. All GRPO experiments use `max_completion_length`=256, resulting in 80–84% truncation of multi-turn responses. This setting is shared across all conditions and does not affect relative comparisons; we retain it because response-level decisions typically manifest within the first 100–200 tokens.

Hyperparameters. Table 5 compares training configurations for GRPO variants and ACT.

Table 5: Training hyperparameters. All experiments use Qwen3-8B, DeepSpeed ZeRO-2, bf16 precision.

	GRPO (F1, F3)[e]	ACT
Learning rate	5e-5	5e-5
Effective batch	16	16
Max steps	8,000	8,000
Warmup steps	200	200
LR schedule	linear[a]	linear
Grad. clipping	—	1.0
LoRA rank	64	64
LoRA α	64	128
LoRA modules	4[b]	7[c]
LoRA dropout	0	0.05
Group size	2[d]	—
KL β	0.04	—
Max compl. len.	256	—
Max seq. len.	—	1,024
L2 loss weight	—	0.1

[a]Cosine schedule for R_3-only. [b]q/k/v/o_proj. [c]+ gate/up/down_proj ($\sim$1.75$\times$ params; §6). [d]Below typical 8–64 (Shao et al., 2024); not tuned. [e]Binary probe (F2) uses a custom SFT training loop rather than GRPO; group size is N/A. LoRA and optimizer settings match the GRPO column.

GRPO reward variants. Cosine R_1 (F1d): w_1=0.40, scale=5.0, w_3=0.60; reward computed at layers [19, 20, 24, 27, 32, 33, 35]; recalibration every 500 steps (§2.3). R_3-only: w_3=1.0; NLI model `cross-encoder/nli-deberta-v3-base`. F1a–F1c weight configurations are reported in Table 2.

B Stubbornness Taxonomy

Manual inspection of all 181 valid-correction responses from the probe-trained model (exp-008v3, step 2000, ToF = 0) reveals five failure categories:

- *Deliberation traps* (28.7%): the model explicitly identifies the correct revised answer

during its reasoning chain but fails to commit, reverting to the original answer in the final output.

- *Surface compliance / sycophantic stubbornness* (24.3%): the model produces agreement markers ("you're right," "good point") but retains its original answer verbatim, simultaneously sycophantic and stubborn.
- *Acknowledge-but-maintain* (23.8%): the model acknowledges the user's correction as plausible but explicitly maintains its original position without substantive engagement.
- *Genuine-accept-missed* (18.8%): the model appears to accept the correction in its reasoning but the final answer format does not reflect the change, suggesting a generation rather than epistemic failure.
- *Refute/ignore/other* (4.5%): the model directly refutes the user's input or ignores it entirely.

The first three categories (76.8%) share a common pattern: the model learns to modulate its *refusal style* in response to evidence strength rather than adjusting its *acceptance rate*, confirming that the probe reward optimizes surface markers rather than epistemic calibration.